

HUADE, China

WMO: 533910

Lat: 41.8956N

Lon: 114.0014E

Elev: 4869

StdP: 12.29

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-9.6	-23.2	1.8	-5.6	-19.0	2.3	-3.4	25.3	-0.6	23.1	0.8	8.2	290	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	84.5	60.2	81.5	59.2	78.7	58.3	66.1	74.5	64.4	73.1	62.8	71.8	8.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
63.6	105.7	69.3	61.7	98.8	68.2	59.8	92.0	67.1	33.7	74.4	32.3	72.7	31.0	71.8	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
22.5	19.6	17.2	DB	-17.5	88.6	4.9	3.3	-21.0	90.9	-23.9	92.9	-26.6	94.7	-30.1	97.1	
			WB	-18.0	68.1	4.8	2.0	-21.4	69.5	-24.2	70.6	-26.9	71.7	-30.4	73.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	39.2	6.2	13.7	27.0	42.0	53.9	63.3	68.1	65.0	55.0	40.8	23.6	10.1	(6)
(7)		DBStd	23.11	8.85	10.05	10.61	9.91	8.45	6.07	4.74	4.93	7.41	8.99	10.32	8.81	(7)
(8)		HDD50	5793	1357	1018	714	274	56	2	0	0	37	304	793	1238	(8)
(9)		HDD65	9662	1822	1438	1178	689	353	100	22	60	305	749	1243	1703	(9)
(10)		CDD50	1847	0	0	1	35	177	400	561	465	188	20	0	0	(10)
(11)		CDD65	243	0	0	0	0	10	48	118	61	6	0	0	0	(11)
(12)		CDH74	2009	0	0	0	9	137	469	905	428	61	0	0	0	(12)
(13)		CDH80	417	0	0	0	1	21	103	219	67	6	0	0	0	(13)
(14)	Wind	WSAvg	7.7	8.0	7.9	8.8	9.3	9.0	6.9	6.0	5.6	6.3	7.4	8.2	8.5	(14)
(15)	Precipitation	PrecAvg	12.6	0.1	0.1	0.2	0.5	1.0	2.2	3.3	3.0	1.4	0.6	0.2	0.1	(15)
(16)		PrecMax	17.5	0.3	0.6	0.4	1.7	2.4	6.4	7.6	6.9	3.1	1.7	0.6	0.3	(16)
(17)		PrecMin	6.5	0.0	0.0	0.0	0.0	0.1	0.6	0.4	0.8	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	2.7	0.1	0.1	0.1	0.4	0.6	1.4	1.7	1.4	0.8	0.4	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	43.3	60.1	74.6	82.9	87.4	89.3	85.8	79.9	69.2	53.1	36.2	(19)
(20)			MCWB	25.2	32.0	40.1	49.3	52.7	58.9	62.8	60.3	55.1	49.3	39.0	27.5	(20)
(21)		2%	DB	27.8	37.4	53.5	68.6	77.9	82.9	85.2	81.6	75.0	63.5	47.0	30.4	(21)
(22)			MCWB	21.8	28.4	37.1	45.8	51.6	57.2	62.2	60.6	54.7	46.9	35.4	24.2	(22)
(23)		5%	DB	23.8	33.4	48.1	63.6	73.9	79.4	82.1	78.6	71.5	59.2	42.7	26.5	(23)
(24)			MCWB	19.0	26.0	34.6	43.3	50.0	56.6	61.2	59.1	52.9	44.1	33.2	21.6	(24)
(25)		10%	DB	19.8	29.4	43.4	58.9	69.9	75.9	79.1	75.8	68.1	55.0	38.2	23.0	(25)
(26)			MCWB	15.9	23.4	31.8	41.3	48.4	55.3	60.5	59.1	51.5	42.0	30.8	18.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.9	32.8	41.9	51.9	56.5	63.3	69.0	67.8	61.6	51.4	39.8	27.9	(27)
(28)			MCD B	32.6	41.1	56.9	67.1	73.0	75.2	79.5	76.1	69.8	63.4	48.5	35.1	(28)
(29)		2%	WB	22.1	29.4	37.9	47.5	54.3	61.5	66.9	65.6	58.2	48.4	36.6	24.3	(29)
(30)			MCD B	27.5	36.0	51.7	64.4	70.7	73.4	76.4	73.4	68.1	59.5	45.7	29.7	(30)
(31)		5%	WB	18.9	26.4	35.1	44.9	52.2	59.9	65.2	63.9	55.9	46.0	33.8	21.8	(31)
(32)			MCD B	23.4	33.2	47.2	59.9	68.3	71.7	74.2	71.4	65.8	55.9	41.8	26.2	(32)
(33)		10%	WB	16.0	23.4	32.3	42.6	50.3	58.3	63.8	62.3	54.0	43.7	31.1	19.0	(33)
(34)			MCD B	19.7	29.2	42.8	56.7	65.1	70.0	73.0	70.1	64.5	52.8	38.1	22.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.0	18.7	19.8	20.5	20.9	19.4	17.9	18.0	19.0	18.5	17.4	16.2	(35)
(36)			MCD BR	21.6	22.3	24.6	26.3	25.6	24.4	22.5	22.3	23.2	23.5	21.5	20.3	(36)
(37)			MCWBR	16.9	15.9	14.5	13.3	11.3	9.0	7.4	8.2	9.8	12.2	13.7	15.7	(37)
(38)		5% WB	MCD BR	21.1	21.6	23.6	23.6	22.4	19.1	17.8	16.7	19.5	20.7	19.9	19.8	(38)
(39)			MCWBR	16.7	15.8	14.6	12.9	10.8	8.1	7.4	7.3	9.7	11.9	13.3	15.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.269	0.315	0.353	0.409	0.452	0.493	0.465	0.452	0.409	0.356	0.299	0.278	(40)	
(41)		taud	2.306	2.178	2.104	2.007	1.937	1.907	2.047	2.055	2.104	2.184	2.275	2.278	(41)	
(42)		Ebn at Noon	280	279	280	272	263	251	257	256	259	259	263	263	(42)	
(43)		Edh at Noon	27	36	45	53	59	61	52	50	44	36	28	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	757	1075	1552	1878	2103	1978	1950	1823	1532	1188	822	653	(44)	
(45)		RadStd	82	86	118	121	134	151	90	97	79	60	79	57	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	-3.31	N/A	N/A	-0.90	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	-2.27	N/A	-0.72	-1.06	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page